电动力学

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符号表

符号	含义
δ_{ij}	Kronecker 符号
$\cdot$	点乘
$\times$	叉乘
$\boldsymbol{a}$	矢量函数
f	标量函数
a_i	矢量 $\boldsymbol{a}$ 的第 i 个分量
δ_{ij}	克罗内克尔符号
$\delta(\boldsymbol{x})$	狄拉克函数
ε_{ijk}	Levi–Civita 符号（三维）
ε_0	真空介电常数
x_i	第 i 个坐标分量
$\boldsymbol{e}$	单位矢量
∇	哈密顿算子
r	$\sqrt{x_i x_i}$
$\boldsymbol{r}$	$x_i \boldsymbol{e}_i$
$\boldsymbol{R}$	旋转矩阵
$\boldsymbol{\Phi}_E$	电场强度通量
ρ	电荷密度

1 矢量分析

1.1 定义

1.1.1 爱因斯坦求和约定

$$\sum_{i=1}^{3} a_i b_i \equiv a_i b_i \tag{1.1}$$

1.1.2 克罗内克尔符号

$$\delta_{ij} \equiv \begin{cases} 1 \ (i = j) \\ 0 \ (i \neq j) \end{cases} \tag{1.2}$$

性质:

$$a_i b_j \delta_{ij} = a_i b_i \tag{1.3}$$

$$\delta_{ij} \delta_{jk} = \delta_{ik} \tag{1.4}$$

1.1.3 点乘

$$\boldsymbol{a} \cdot \boldsymbol{b} \equiv a_i b_i \tag{1.5}$$

1.1.4 Levi-Civita 符号

$$\varepsilon_{ijk} \equiv \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} \tag{1.6}$$

性质:

$$\varepsilon_{ijk} = -\varepsilon_{ikj} = -\varepsilon_{kji} = -\varepsilon_{jik} \tag{1.7}$$

$$
\begin{aligned}
\varepsilon_{ijk}\varepsilon_{lmn} &= \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix}^T \begin{pmatrix} \delta_{1l} & \delta_{2l} & \delta_{3l} \\ \delta_{1m} & \delta_{2m} & \delta_{3m} \\ \delta_{1n} & \delta_{2n} & \delta_{3n} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{1i} & \delta_{1j} & \delta_{1k} \\ \delta_{2i} & \delta_{2j} & \delta_{2k} \\ \delta_{3i} & \delta_{3j} & \delta_{3k} \end{pmatrix} \begin{pmatrix} \delta_{1l} & \delta_{2l} & \delta_{3l} \\ \delta_{1m} & \delta_{2m} & \delta_{3m} \\ \delta_{1n} & \delta_{2n} & \delta_{3n} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{1l} & \delta_{2l} & \delta_{3l} \\ \delta_{1m} & \delta_{2m} & \delta_{3m} \\ \delta_{1n} & \delta_{2n} & \delta_{3n} \end{pmatrix} \begin{pmatrix} \delta_{1i} & \delta_{1j} & \delta_{1k} \\ \delta_{2i} & \delta_{2j} & \delta_{2k} \\ \delta_{3i} & \delta_{3j} & \delta_{3k} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{li} & \delta_{lj} & \delta_{lk} \\ \delta_{mi} & \delta_{mj} & \delta_{mk} \\ \delta_{ni} & \delta_{nj} & \delta_{nk} \end{pmatrix}
\end{aligned} \tag{1.8}
$$

$$
\varepsilon_{ijk}\varepsilon_{lmk} = \sum_{k=1}^{3} \begin{pmatrix} \delta_{li} & \delta_{lj} & \delta_{lk} \\ \delta_{mi} & \delta_{mj} & \delta_{mk} \\ \delta_{ki} & \delta_{kj} & 1 \end{pmatrix} = \begin{pmatrix} \delta_{li} & \delta_{lj} & 0 \\ \delta_{mi} & \delta_{mj} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \delta_{li}\delta_{mj} - \delta_{lj}\delta_{mi} \tag{1.9}
$$

$$
\varepsilon_{ijk}\varepsilon_{ljk} = \delta_{li}\delta_{jj} - \delta_{lj}\delta_{ji} = 3\delta_{li} - \delta_{li} = 2\delta_{li} \tag{1.10}
$$

1.1.5 叉乘

$$
\boldsymbol{a} \times \boldsymbol{b} \equiv \varepsilon_{ijk} a_i b_j \boldsymbol{e}_k = \begin{pmatrix} \boldsymbol{e}_i & \boldsymbol{e}_j & \boldsymbol{e}_k \\ a_i & a_j & a_k \\ b_i & b_j & b_k \end{pmatrix} \tag{1.11}
$$

性质:

$$
\begin{aligned}
\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b}) &= \boldsymbol{c}\cdot\varepsilon_{ijk}a_ib_j\boldsymbol{e}_k \\
&= c_k\varepsilon_{ijk}a_ib_j \\
&= \varepsilon_{ijk}a_ib_jc_k \\
&= \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} a_ib_jc_k \\
&= \begin{pmatrix} \delta_{1i}a_i & \delta_{2i}a_i & \delta_{3i}a_i \\ \delta_{1j}b_j & \delta_{2j}b_j & \delta_{3j}b_j \\ \delta_{1k}c_k & \delta_{2k}c_k & \delta_{3k}c_k \end{pmatrix} \\
&= \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}
\end{aligned} \tag{1.12}
$$

$$
\begin{aligned}
\boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b}) &= \boldsymbol{c}\times\varepsilon_{ijk}a_ib_j\boldsymbol{e}_k \\
&= \varepsilon_{lkn}c_l\varepsilon_{ijk}a_ib_j\boldsymbol{e}_n \\
&= -\varepsilon_{lnk}\varepsilon_{ijk}a_ib_jc_l\boldsymbol{e}_n \\
&= -(\delta_{il}\delta_{jn}-\delta_{in}\delta_{jl})a_ib_jc_l\boldsymbol{e}_n \\
&= -\delta_{il}\delta_{jn}a_ib_jc_l\boldsymbol{e}_n+\delta_{in}\delta_{jl}a_ib_jc_l\boldsymbol{e}_n \\
&= -a_ib_jc_i\boldsymbol{e}_j+a_ib_jc_j\boldsymbol{e}_i \\
&= -(\boldsymbol{a}\cdot\boldsymbol{c})\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{c})\boldsymbol{a}
\end{aligned} \tag{1.13}
$$

$$
\boldsymbol{e}\times(\boldsymbol{e}\times\boldsymbol{b}) = -(\boldsymbol{e}\cdot\boldsymbol{e})\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{e})\boldsymbol{e} = -\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{e})\boldsymbol{e} \tag{1.14}
$$

1.1.6 哈密顿算子

$$
\boldsymbol{\nabla} \equiv \frac{\partial}{\partial x_i}\boldsymbol{e}_i \tag{1.15}
$$

1.1.7 梯度

设位置矢量 $\boldsymbol{r}=\boldsymbol{r}(u_1,u_2,u_3)$, 拉梅系数为

$$
h_i \equiv \left\|\frac{\partial\boldsymbol{r}}{\partial u_i}\right\|, \qquad \boldsymbol{e} \equiv \frac{1}{h_i}\frac{\partial\boldsymbol{r}}{\partial u_i}
$$

且 $\boldsymbol{e}_i\cdot\boldsymbol{e}_j=\delta_{ij}$。

对标量场 $f(u_1,u_2,u_3)$, 全微分为

$$
\mathrm{d}f = \frac{\partial f}{\partial u_i}\,\mathrm{d}u_i.
$$

无穷小位移（线元）为

$$\mathrm{d}\boldsymbol{l} = h_i\ \mathrm{d}u_i\,\boldsymbol{e}_i.$$

设 $\boldsymbol{\nabla} f = G_i\boldsymbol{e}_i$, 则

$$\mathrm{d}f = \boldsymbol{\nabla} f\cdot\boldsymbol{dl} = G_i h_i\ \mathrm{d}u_i.$$

比较系数, 得 $\boldsymbol{G} = \dfrac{1}{h_i}\dfrac{\partial f}{\partial u_i}\boldsymbol{e}_i$, 从而

$$\boldsymbol{\nabla} f = \frac{1}{h_i}\frac{\partial f}{\partial u_i}\boldsymbol{e}_i \tag{1.16}$$

对于直角坐标

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial x_i}\boldsymbol{e}_i \tag{1.17}$$

性质:

$$\boldsymbol{\nabla} r = \frac{\partial\sqrt{x_i x_i}}{\partial x_i}\boldsymbol{e}_i = \frac{x_i}{r}\boldsymbol{e}_i = \frac{\boldsymbol{r}}{r} \tag{1.18}$$

$$\boldsymbol{\nabla} f(r) = \frac{\partial f}{\partial r}\frac{\partial r}{\partial x_i}\boldsymbol{e}_i = \frac{\partial f}{\partial r}\frac{\boldsymbol{r}}{r} \tag{1.19}$$

1.1.8 散度

对微小增量 $\mathrm{d}u_1, \mathrm{d}u_2, \mathrm{d}u_3$, 线元在三方向的实际长度分别为

$$\mathrm{d}l_1 = h_1\,\mathrm{d}u_1,\quad \mathrm{d}l_2 = h_2\,\mathrm{d}u_2,\quad \mathrm{d}l_3 = h_3\,\mathrm{d}u_3$$

因此微小长方体的体元为

$$\mathrm{d}\tau = \mathrm{d}l_1\,\mathrm{d}l_2\,\mathrm{d}l_3 = h_1h_2h_3\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3. \tag{1.20}$$

设矢量场在该点为

$$\boldsymbol{a} = a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + a_3\boldsymbol{e}_3,$$

其中 a_1, a_2, a_3 为在局部单位基矢下的分量（均为 u_1, u_2, u_3 的函数）。

下面计算穿过微小长方体六个面的净通量。以 u_1 方向为例, 考察在 u_1 与 $u_1 + \mathrm{d}u_1$ 两个面：

面 u_1（在坐标 u 处, 向外法向为 $-\boldsymbol{e}_1$）：面积元为

$$\mathrm{d}\boldsymbol{a}_1^{(-)} = -\boldsymbol{e}_1(h_2\,\mathrm{d}u_2)(h_3\,\mathrm{d}u_3) = -\boldsymbol{e}_1h_2h_3\,\mathrm{d}u_2\,\mathrm{d}u_3,$$

对应的通量（近似取该面中心处分量）：

$$\Phi_1^{(-)} = \boldsymbol{a}(u_1,u_2,u_3)\cdot\mathrm{d}\boldsymbol{a}_1^{(-)} = -a_1(u_1,u_2,u_3)h_2h_3\,\mathrm{d}u_2\,\mathrm{d}u_3.$$

面 $u + \mathrm{d}u$（向外法向为 $+\boldsymbol{e}_1$）：面积元

$$\mathrm{d}\boldsymbol{a}_1^{(+)} = +\boldsymbol{e}_1h_2h_3\,\mathrm{d}u_2\,\mathrm{d}u_3,$$

通量（在 $u + du$ 处）：

$$\Phi_1^{(+)} = a_1(u_1+du_1,u_2,u_3)h_2(u_1+du_1,u_2,u_3)h_3(u_1+du_1,u_2,u_3)\,\mathrm{d}u_2\,\mathrm{d}u_3.$$

因此穿过这对面的净通量为

$$\begin{aligned}\Delta\Phi_1 &= \Phi_1^{(+)} - \Phi_1^{(-)}\\ &= [a_1(u_1+\mathrm{d}u_1,u_2,u_3)h_2(u_1+\mathrm{d}u_1,u_2,u_3)h_3(u_1+\mathrm{d}u_1,u_2,u_3) - a_1(u_1,u_2,u_3)h_2h_3]\,\mathrm{d}u_2\,\mathrm{d}u_3.\end{aligned}$$

用泰勒展开到一阶并忽略高阶项, 得

$$\Delta\Phi_u = \frac{\partial}{\partial u_1}(h_2h_3a_1)\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3 + o(\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3).$$

对 u_2 和 u_3 方向做同样的计算, 六个面的净通量近似为三项之和：

$$\Delta\Phi_{\text{total}} = \frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right)\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3$$

由于体元按照式(1.20)为 $\mathrm{d}\tau = h_1h_2h_3\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3$, 故单位体积的通量密度（即散度）为极限：

$$\begin{aligned}\boldsymbol{\nabla}\cdot\boldsymbol{a} &= \lim_{\Delta\tau\to 0}\frac{\Delta\Phi_{\text{total}}}{d\tau}\\ &= \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right).\end{aligned}$$

于是得到正交曲线坐标系下的散度公式：

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right). \tag{1.21}$$

对于直角坐标

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = \frac{\partial a_i}{\partial x_i} \tag{1.22}$$

性质:

$$\boldsymbol{\nabla}\cdot\boldsymbol{r} = 3 \tag{1.23}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{f}(r) = \frac{\partial f_i(r)}{\partial x_i} = f_i'\frac{\partial r}{\partial x_i} = \boldsymbol{f}'\cdot\frac{\boldsymbol{r}}{r} \tag{1.24}$$

1.1.9 旋度

选取微小矩形位于 u_1-u_2 平面, 固定 u_3。矩形四边的线元素：

$$\begin{aligned}
&\text{下边: } \mathrm{d}\boldsymbol{l}_1 = h_1\,\mathrm{d}u_1\boldsymbol{e}_1, \quad \text{从}(u_1,u_2)\to(u_1+\mathrm{d}u_1,u_2)\\
&\text{右边: } \mathrm{d}\boldsymbol{l}_2 = h_2\,\mathrm{d}u_2\boldsymbol{e}_2, \quad \text{从}(u_1+\mathrm{d}u_1,u_2)\to(u_1+\mathrm{d}u_1,u_2+\mathrm{d}u_2)\\
&\text{上边: } \mathrm{d}\boldsymbol{l}_3 = -h_1\,\mathrm{d}u_1\boldsymbol{e}_1, \quad \text{从}(u_1+\mathrm{d}u_1,u_2+\mathrm{d}u_2)\to(u_1,u_2+\mathrm{d}u_2)\\
&\text{左边: } \mathrm{d}\boldsymbol{l}_4 = -h_2du_2\boldsymbol{e}_2, \quad \text{从}(u_1,u_2+\mathrm{d}u_2)\to(u_1,u_2)
\end{aligned}$$

沿每一边计算 $\boldsymbol{a}\cdot d\boldsymbol{l}$ 并做泰勒展开（只保留一阶项）：

$$\oint\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l} \approx \left[\frac{\partial}{\partial u_1}(h_2a_2) - \frac{\partial}{\partial u_2}(h_1a_1)\right]\mathrm{d}u_1\,\mathrm{d}u_2$$

面元面积：

$$dS_3 = h_1h_2\,\mathrm{d}u_1\,\mathrm{d}u_2$$

旋度在 $\boldsymbol{e}_3$ 方向的分量：

$$(\boldsymbol{\nabla}\times\boldsymbol{a})_3 = \lim_{\mathrm{d}u_1,\mathrm{d}u_2\to 0}\frac{\oint\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}}{\mathrm{d}S_3} = \frac{1}{h_1h_2}\left[\frac{\partial}{\partial u_1}(h_2a_2) - \frac{\partial}{\partial u_2}(h_1a_1)\right]$$

同理, 对于其他两个方向：

$$(\boldsymbol{\nabla}\times\boldsymbol{a})_1 = \frac{1}{h_2h_3}\left[\frac{\partial}{\partial u_2}(h_3a_3) - \frac{\partial}{\partial u_3}(h_2a_2)\right], \quad (\boldsymbol{\nabla}\times\boldsymbol{a})_2 = \frac{1}{h_3h_1}\left[\frac{\partial}{\partial u_3}(h_1a_1) - \frac{\partial}{\partial u_1}(h_3a_3)\right]$$

最终正交曲线坐标系下旋度公式为：

$$\boldsymbol{\nabla} \times \boldsymbol{a} = \frac{h_i}{h_1 h_2 h_3} \varepsilon_{ijk} \frac{\partial (h_k a_k)}{\partial u_j} \boldsymbol{e}_i \tag{1.25}$$

对于直角坐标

$$\boldsymbol{\nabla} \times \boldsymbol{a} = \varepsilon_{ijk} \frac{\partial a_j}{\partial x_i} \boldsymbol{e}_k \tag{1.26}$$

性质:

$$\boldsymbol{\nabla} \times \boldsymbol{r} = 0 \tag{1.27}$$

$$\boldsymbol{\nabla} \times (f\boldsymbol{a}) = \varepsilon_{ijk} \frac{\partial (f a_j)}{\partial x_i} \boldsymbol{e}_k = f \varepsilon_{ijk} \frac{\partial a_j}{\partial x_i} \boldsymbol{e}_k + \varepsilon_{ijk} \frac{\partial f}{\partial x_i} a_j \boldsymbol{e}_k = f(\boldsymbol{\nabla} \times \boldsymbol{a}) + (\boldsymbol{\nabla} f) \times \boldsymbol{a} \tag{1.28}$$

1.1.10 并积

$$\boldsymbol{\nabla} \boldsymbol{a} \equiv \begin{pmatrix} \frac{\partial a_1}{x_1} & \frac{\partial a_2}{x_1} & \frac{\partial a_3}{x_1} \\ \frac{\partial a_1}{x_2} & \frac{\partial a_2}{x_2} & \frac{\partial a_3}{x_2} \\ \frac{\partial a_1}{x_3} & \frac{\partial a_2}{x_3} & \frac{\partial a_3}{x_3} \end{pmatrix} = \frac{\partial a_i}{\partial x_j} \boldsymbol{e}_i \boldsymbol{e}_j \tag{1.29}$$

1.1.11 拉普拉斯算子

定义为梯度的散度, 由1.16和1.21得

$$\begin{aligned} \boldsymbol{\nabla}^2 f &= \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} f) \\ &= \boldsymbol{\nabla} \cdot \left(\frac{1}{h_i} \frac{\partial f}{\partial u_i} \right) \\ &= \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial u_i} \left(h_1 h_2 h_3 \frac{\frac{1}{h_i} \frac{\partial f}{\partial u_i}}{h_i} \right) \\ &= \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial u_i} \left(h_1 h_2 h_3 \frac{\partial f}{h_i^2 \partial u_i} \right) \end{aligned} \tag{1.30}$$

对于直角坐标

$$\boldsymbol{\nabla}^2 = \frac{\partial^2}{\partial x_i^2} \tag{1.31}$$

性质:

$$
\begin{aligned}
\boldsymbol{\nabla}^2(fg) &= \frac{\partial}{\partial x_i}\frac{\partial fg}{\partial x_i} \\
&= \frac{\partial}{\partial x_i}\left(g\frac{\partial f}{\partial x_i} + f\frac{\partial g}{\partial x_i}\right) \\
&= \frac{\partial}{\partial x_i}\left(g\frac{\partial f}{\partial x_i}\right) + \frac{\partial}{\partial x_i}\left(f\frac{\partial g}{\partial x_i}\right) \\
&= \frac{\partial}{\partial x_i}\left(g\frac{\partial f}{\partial x_i}\right) + \frac{\partial}{\partial x_i}\left(f\frac{\partial g}{\partial x_i}\right) \\
&= \frac{\partial g}{\partial x_i}\frac{\partial f}{\partial x_i} + g\frac{\partial^2 f}{\partial x_i^2} + \frac{\partial f}{\partial x_i}\frac{\partial g}{\partial x_i} + a\frac{\partial^2 g}{\partial x_i^2} \\
&= g\frac{\partial^2 f}{\partial x_i^2} + 2\frac{\partial f}{\partial x_i}\frac{\partial g}{\partial x_i} + f\frac{\partial^2 g}{\partial x_i^2} \\
&= g\boldsymbol{\nabla}^2 f + 2(\boldsymbol{\nabla} f)\cdot(\boldsymbol{\nabla} g) + f\boldsymbol{\nabla}^2 g
\end{aligned} \tag{1.32}
$$

1.2 微分运算

1.2.1 和规则

$$\boldsymbol{\nabla}(f+g) = \boldsymbol{\nabla} f + \boldsymbol{\nabla} g \tag{1.33}$$

$$\boldsymbol{\nabla}\cdot(\boldsymbol{a}+\boldsymbol{b}) = \boldsymbol{\nabla}\cdot\boldsymbol{a} + \boldsymbol{\nabla}\cdot\boldsymbol{b} \tag{1.34}$$

$$\boldsymbol{\nabla}\times(\boldsymbol{a}+\boldsymbol{b}) = \boldsymbol{\nabla}\times\boldsymbol{a} + \boldsymbol{\nabla}\times\boldsymbol{b} \tag{1.35}$$

1.2.2 积规则

$$\boldsymbol{\nabla}(fg) = g\boldsymbol{\nabla} f + f\boldsymbol{\nabla} g \tag{1.36}$$

$$
\begin{aligned}
&\boldsymbol{a}\times(\boldsymbol{\nabla}\times\boldsymbol{b}) + \boldsymbol{b}\times(\boldsymbol{\nabla}\times\boldsymbol{a}) + (\boldsymbol{a}\cdot\boldsymbol{\nabla})\boldsymbol{b} + (\boldsymbol{b}\cdot\boldsymbol{\nabla})\boldsymbol{a} \\
=&\boldsymbol{a}\times\left(\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_k\right) + \boldsymbol{b}\times\left(\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_k\right) + \left(a_i\frac{\partial}{\partial x_i}\right)\boldsymbol{b} + \left(b_i\frac{\partial}{\partial x_i}\right)\boldsymbol{a} \\
=&\varepsilon_{lkn}a_l\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n + \varepsilon_{lkn}b_l\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&-\varepsilon_{lnk}\varepsilon_{ijk}a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n - \varepsilon_{lnk}\varepsilon_{ijk}b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&-(\delta_{il}\delta_{jn} - \delta_{in}\delta_{jl})a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n - (\delta_{il}\delta_{jn} - \delta_{in}\delta_{jl})b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&\left(-\delta_{il}\delta_{jn}a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n + \delta_{in}\delta_{jl}a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n\right) + \left(-\delta_{il}\delta_{jn}b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + \delta_{in}\delta_{jl}b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n\right) + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&\left(-a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + a_j\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_i\right) + \left(-b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j + b_j\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_i\right) + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&a_j\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_i + b_j\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_i \\
=&\frac{\partial(a_j b_j)}{\partial x_i}\boldsymbol{e}_i \\
=&\boldsymbol{\nabla}(\boldsymbol{a}\cdot\boldsymbol{b})
\end{aligned} \tag{1.37}
$$

$$\boldsymbol{\nabla}\cdot(f\boldsymbol{a})=\frac{\partial f a_i}{\partial x_i}=f\frac{\partial a_i}{\partial x_i}+a_i\frac{\partial f}{\partial x_i}=f\boldsymbol{\nabla}\cdot\boldsymbol{a}+\boldsymbol{a}\cdot\boldsymbol{\nabla}f \tag{1.38}$$

$$\begin{aligned}\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})-\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})&=\boldsymbol{b}\cdot(\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_k)-\boldsymbol{a}\cdot(\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_k)\\&=b_k\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}-a_k\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\\&=b_j\varepsilon_{ijk}\frac{\partial a_i}{\partial x_k}+a_i\varepsilon_{ijk}\frac{\partial b_j}{\partial x_k}\\&=\frac{\partial\varepsilon_{ijk}a_ib_j}{\partial x_k}\\&=\boldsymbol{\nabla}\cdot(\varepsilon_{ijk}a_ib_j\boldsymbol{e}_k)\\&=\boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{b})\end{aligned} \tag{1.39}$$

1.2.3 二阶积规则

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}f)=\boldsymbol{\nabla}\times(\frac{\partial f}{\partial x_i}\boldsymbol{e}_i)=\varepsilon_{ijk}\frac{\partial}{\partial x_i}\frac{\partial f}{\partial x_j}\boldsymbol{e}_k=0 \tag{1.40}$$

$$\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})=\boldsymbol{\nabla}\cdot(\varepsilon_{ijk}\frac{\partial a_i}{\partial x_j}\boldsymbol{e}_k)=\varepsilon_{ijk}\frac{\partial^2 a_i}{\partial x_j\partial x_k}=0 \tag{1.41}$$

$$\Delta f=\frac{\partial^2 f}{\partial x_i^2} \tag{1.42}$$

$$\Delta\boldsymbol{a}=\frac{\partial^2(a_j\boldsymbol{e}_j)}{\partial x_i^2} \tag{1.43}$$

$$\begin{aligned}\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{a})&=\boldsymbol{\nabla}\times(\varepsilon_{ijk}\frac{\partial a_i}{\partial x_j}\boldsymbol{e}_k)\\&=\varepsilon_{lmn}\varepsilon_{ijl}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=\varepsilon_{mnl}\varepsilon_{ijl}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=(\delta_{im}\delta_{jn}-\delta_{in}\delta_{jm})\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=\delta_{im}\delta_{jn}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n-\delta_{in}\delta_{jm}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=\frac{\partial^2 a_i}{\partial x_j\partial x_i}\boldsymbol{e}_j-\frac{\partial^2 a_i}{\partial x_j\partial x_j}\boldsymbol{e}_i\\&=\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{a})-\Delta\boldsymbol{a}\end{aligned} \tag{1.44}$$

1.3 积分运算

1.3.1 高斯定理

我们已经得到正交坐标系中的散度局部公式1.21。因此, 对体积元 $\Delta\tau=h_1h_2h_3\,\Delta u_1\Delta u_2\Delta u_3$ 的六个面计算净通量时：

$$\Phi^{(\mathrm{out})}-\Phi^{(\mathrm{in})}=\left[\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right)\right]\Delta u_1\Delta u_2\Delta u_3=(\boldsymbol{\nabla}\cdot\boldsymbol{a})\,\Delta\tau.$$

将整个区域 V 分割成许多此类体积元, 内部公共面的通量相互抵消, 仅留下外边界 ∂V 的通量。令分割尺寸趋于零, 通量和体积和分别趋于各自积分, 故得到

$$\iiint_V (\boldsymbol{\nabla}\cdot\boldsymbol{a})\,\mathrm{d}V = \iint_{\partial V} \boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S} \tag{1.45}$$

1.3.2 斯托克斯定理

我们已经得到任意正交坐标系中的旋度局部公式1.25 , 考虑由 (u_1, u_2) 增量生成的一个无穷小曲面元

$$\mathrm{d}\boldsymbol{S} = (h_1 h_2\,\mathrm{d}u_1\,\mathrm{d}u_2)\boldsymbol{e}_3.$$

沿此小曲面元的边界进行线积分, 可得

$$\oint\!\!\!\oint_{\partial S} \boldsymbol{a}\cdot d\boldsymbol{l} = (\nabla\times\boldsymbol{a})\cdot d\boldsymbol{S}.$$

将整个曲面分割成许多这样的小元, 内部公共边界的线积分相互抵消, 仅留下外边界 ∂S 的线积分。令分割尺寸趋于零, 总和趋于积分, 故得

$$\iint_S (\boldsymbol{\nabla}\times\boldsymbol{a})\cdot\mathrm{d}\boldsymbol{S} = \oint\!\!\!\oint_{\partial S} \boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l} \tag{1.46}$$

1.4 曲面坐标系

1.4.1 柱坐标系

将柱坐标系坐标转化为直角系坐标的方式为

$$x_1 = s\cos\phi,\ x_2 = s\sin\phi,\ x_3 = x_3 \tag{1.47}$$

其中 s 是坐标点到 z 轴的距离。柱坐标系和直角系坐标无限小元转换方式为

$$\begin{pmatrix}\mathrm{d}x_1\\ \mathrm{d}x_2\\ \mathrm{d}x_3\end{pmatrix} = \begin{pmatrix}\frac{x_1}{s} & -x_2 & 0\\ \frac{x_2}{s} & x_1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\mathrm{d}s\\ \mathrm{d}\phi\\ \mathrm{d}x_3\end{pmatrix} \tag{1.48}$$

其拉梅系数为

$$h_1 = 1,\ h_2 = s,\ h_3 = 1 \tag{1.49}$$

其无限小位移为

$$\mathrm{d}\boldsymbol{l} = h_i\,\mathrm{d}u_i\boldsymbol{e}_i = \mathrm{d}s\boldsymbol{s} + s\,\mathrm{d}\phi\boldsymbol{\phi} + \mathrm{d}x_3\boldsymbol{x}_3 \tag{1.50}$$

其体积元为

$$\mathrm{d}V = s\,\mathrm{d}s\,\mathrm{d}\phi\,\mathrm{d}x_3 \tag{1.51}$$

梯度

$$\nabla f = \frac{1}{h_i}\frac{\partial f}{\partial u_i}\boldsymbol{e}_i = \frac{\partial}{\partial s}\boldsymbol{s} + \frac{1}{s}\frac{\partial}{\partial\phi}\boldsymbol{\phi} + \frac{\partial}{\partial x_3}\boldsymbol{x}_3 \tag{1.52}$$

散度

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = \frac{1}{s}(\frac{\partial s a_s}{\partial s} + \frac{\partial a_\phi}{\partial\phi} + \frac{\partial s a_3}{\partial x_3}) = \frac{1}{s}\frac{\partial s a_s}{\partial s} + \frac{1}{s}\frac{\partial a_\phi}{\partial\phi} + \frac{\partial a_3}{\partial x_3} = \frac{a_s}{s} + \frac{\partial a_s}{\partial s} + \frac{1}{s}\frac{\partial a_\phi}{\partial\phi} + \frac{\partial a_3}{\partial x_3} \tag{1.53}$$

旋度

$$
\begin{aligned}
\boldsymbol{\nabla}\times\boldsymbol{a} &= \frac{h_i}{h_1h_2h_3}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i \\
&= \frac{h_i}{s}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i \\
&= \frac{1}{s}\left(\varepsilon_{123}\frac{\partial a_3}{\partial u_2}\boldsymbol{e}_1+\varepsilon_{132}\frac{\partial sa_2}{\partial u_3}\boldsymbol{e}_1+\varepsilon_{321}\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3+s\varepsilon_{213}\frac{\partial a_3}{\partial u_1}\boldsymbol{e}_2+\varepsilon_{312}\frac{\partial sa_2}{\partial u_1}\boldsymbol{e}_3+s\varepsilon_{231}\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right) \\
&= \frac{1}{s}\left(\frac{\partial a_3}{\partial u_2}\boldsymbol{e}_1-\frac{\partial sa_2}{\partial u_3}\boldsymbol{e}_1-\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3-s\frac{\partial a_3}{\partial u_1}\boldsymbol{e}_2+\frac{\partial sa_2}{\partial u_1}\boldsymbol{e}_3+s\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right) \\
&= \frac{1}{s}\left(\frac{\partial a_3}{\partial\phi}\boldsymbol{s}-\frac{\partial sa_\phi}{\partial x_3}\boldsymbol{s}-\frac{\partial a_s}{\partial\phi}\boldsymbol{e}_3-s\frac{\partial a_3}{\partial s}\boldsymbol{\phi}+\frac{\partial sa_\phi}{\partial s}\boldsymbol{e}_3+s\frac{\partial a_s}{\partial x_3}\boldsymbol{\phi}\right) \\
&= \frac{1}{s}\left(\frac{\partial a_3}{\partial\phi}-s\frac{\partial a_\phi}{\partial x_3}\right)\boldsymbol{s}+\left(\frac{\partial a_s}{\partial x_3}-\frac{\partial a_3}{\partial s}\right)\boldsymbol{\phi}+\frac{1}{s}\left(\frac{\partial sa_\phi}{\partial s}-\frac{\partial a_s}{\partial\phi}\right)\boldsymbol{e}_3
\end{aligned}
\tag{1.54}
$$

拉普拉斯算子

$$
\begin{aligned}
\boldsymbol{\nabla}^2 f &= \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{\partial f}{h_i^2\partial u_i}\right) \\
&= \frac{1}{s}\frac{\partial}{\partial u_i}\left(s\frac{\partial f}{h_i^2\partial u_i}\right) \\
&= \frac{1}{s}\frac{\partial}{\partial u_s}(s\frac{\partial f}{h_s^2\partial u_s})+\frac{1}{s}\frac{\partial}{\partial\phi}(s\frac{\partial f}{h_\phi^2\partial\phi})+\frac{1}{s}\frac{\partial}{\partial u_3}(s\frac{\partial f}{h_3^2\partial u_3}) \\
&= \frac{1}{s}\frac{\partial}{\partial u_s}(s\frac{\partial f}{\partial u_s})+\frac{1}{s}\frac{\partial}{\partial\phi}(s\frac{\partial f}{s^2\partial\phi})+\frac{1}{s}\frac{\partial}{\partial u_3}(s\frac{\partial f}{\partial u_3}) \\
&= \frac{1}{s}\frac{\partial}{\partial u_s}(s\frac{\partial f}{\partial u_s})+\frac{1}{s^2}\frac{\partial^2 f}{\partial\phi^2}+\frac{\partial^2 f}{\partial u_3^2}
\end{aligned}
\tag{1.55}
$$

1.4.2 球坐标系

将球坐标系坐标转化为直角系坐标的方式为

$$
x_1 = r\cos\phi\sin\theta,\ x_2 = r\sin\phi\sin\theta,\ x_3 = r\cos\theta \tag{1.56}
$$

球坐标系和直角系坐标无限小元转换方式为

$$
\begin{pmatrix}\mathrm{d}x_1\\ \mathrm{d}x_2\\ \mathrm{d}x_3\end{pmatrix}=\begin{pmatrix}\frac{x_1}{r} & -x_2 & \frac{x_1x_3}{s}\\ \frac{x_2}{r} & x_1 & \frac{x_2x_3}{s}\\ \frac{x_3}{r} & 0 & -s\end{pmatrix}\begin{pmatrix}\mathrm{d}r\\ \mathrm{d}\phi\\ \mathrm{d}\theta\end{pmatrix} \tag{1.57}
$$

其拉梅系数为

$$
h_1 = 1,\ h_2 = s,\ h_3 = r \tag{1.58}
$$

其无限小位移为

$$
\mathrm{d}\boldsymbol{l} = \mathrm{d}r\boldsymbol{r} + s\theta\,\mathrm{d}\phi\boldsymbol{\phi} + r\,\mathrm{d}\theta\boldsymbol{\theta} \tag{1.59}
$$

其体积元为

$$
\mathrm{d}V = sr\,\mathrm{d}r\,\mathrm{d}\phi\,\mathrm{d}\theta \tag{1.60}
$$

梯度

$$
\nabla f = \frac{1}{h_i}\frac{\partial f}{\partial u_i}\boldsymbol{e}_i = \frac{\partial}{\partial r}\boldsymbol{r}+\frac{1}{s}\frac{\partial}{\partial\phi}\boldsymbol{\phi}+\frac{1}{r}\frac{\partial}{\partial\theta}\boldsymbol{\theta} \tag{1.61}
$$

散度

$$\boldsymbol{\nabla}\cdot\boldsymbol{a}=\frac{1}{sr}(\frac{\partial sra_r}{\partial r}+\frac{\partial ra_\phi}{\partial\phi}+\frac{\partial s\theta}{\partial\theta})=\frac{1}{sr}(\frac{\partial sra_r}{\partial r}+r\frac{\partial a_\phi}{\partial\phi}+s\frac{\partial\theta}{\partial\theta})=\frac{1}{sr}\frac{\partial sra_r}{\partial r}+\frac{1}{s}\frac{\partial a_\phi}{\partial\phi}+\frac{1}{r}\frac{\partial\theta}{\partial\theta} \tag{1.62}$$

旋度

$$\begin{aligned}\boldsymbol{\nabla}\times\boldsymbol{a}&=\frac{h_i}{h_1h_2h_3}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i\\&=\frac{h_i}{sr}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i\\&=\frac{1}{sr}\left(\varepsilon_{123}\frac{\partial ra_3}{\partial u_2}\boldsymbol{e}_1+\varepsilon_{132}\frac{\partial sa_2}{\partial u_3}\boldsymbol{e}_1+r\varepsilon_{321}\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3+s\varepsilon_{213}\frac{\partial ra_3}{\partial u_1}\boldsymbol{e}_2+r\varepsilon_{312}\frac{\partial sa_2}{\partial u_1}\boldsymbol{e}_3+s\varepsilon_{231}\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right)\\&=\frac{1}{sr}\left(\frac{\partial ra_3}{\partial u_2}\boldsymbol{e}_1-\frac{\partial sa_2}{\partial u_3}\boldsymbol{e}_1-r\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3-s\frac{\partial ra_3}{\partial u_1}\boldsymbol{e}_2+r\frac{\partial sa_2}{\partial u_1}\boldsymbol{e}_3+s\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right)\\&=\frac{1}{sr}\left(\frac{\partial ra_\theta}{\partial\phi}\boldsymbol{r}-\frac{\partial sa_\phi}{\partial\theta}\boldsymbol{r}-r\frac{\partial a_r}{\partial\phi}\boldsymbol{\theta}-s\frac{\partial ra_\theta}{\partial r}\boldsymbol{\phi}+r\frac{\partial sa_\phi}{\partial r}\boldsymbol{\theta}+s\frac{\partial a_r}{\partial\theta}\boldsymbol{\phi}\right)\\&=\frac{1}{sr}\left(r\frac{\partial a_\theta}{\partial\phi}-s\frac{\partial a_\phi}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial a_r}{\partial\theta}-\frac{\partial ra_\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{s}\left(\frac{\partial sa_\phi}{\partial r}-\frac{\partial a_r}{\partial\phi}\right)\boldsymbol{\theta}\end{aligned} \tag{1.63}$$

拉普拉斯算子

$$\begin{aligned}\boldsymbol{\nabla}^2f&=\frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{\partial f}{h_i^2\partial u_i}\right)\\&=\frac{1}{sr}\frac{\partial}{\partial u_i}\left(sr\frac{\partial f}{h_i^2\partial u_i}\right)\\&=\frac{1}{sr}\frac{\partial}{\partial r}\left(sr\frac{\partial f}{h_r^2\partial r}\right)+\frac{1}{sr}\frac{\partial}{\partial\phi}\left(sr\frac{\partial f}{h_\phi^2\partial\phi}\right)+\frac{1}{sr}\frac{\partial}{\partial\theta}\left(sr\frac{\partial f}{h_\theta^2\partial\theta}\right)\\&=\frac{1}{sr}\frac{\partial}{\partial r}\left(sr\frac{\partial f}{\partial r}\right)+\frac{1}{sr}\frac{\partial}{\partial\phi}\left(r\frac{\partial f}{s\partial\phi}\right)+\frac{1}{sr}\frac{\partial}{\partial\theta}\left(s\frac{\partial f}{r\partial\theta}\right)\\&=\frac{1}{sr}\frac{\partial}{\partial r}\left(sr\frac{\partial f}{\partial r}\right)+\frac{1}{s^2}\frac{\partial}{\partial\phi}\left(\frac{\partial f}{\partial\phi}\right)+\frac{1}{sr^2}\frac{\partial}{\partial\theta}\left(s\frac{\partial f}{\partial\theta}\right)\end{aligned} \tag{1.64}$$

1.5 狄拉克函数与阶跃函数

1.5.1 狄拉克函数

严格来说，狄拉克 δ 函数并不是一个通常意义下的函数，而是通过其在积分中的作用来定义的。设 $f(x)$ 是在 $x=0$ 附近连续，并且在无穷远处足够快衰减的函数。定义狄拉克 δ 满足

$$\int_{-\infty}^{\infty}\delta(x)\,f(x)\,\mathrm{d}x=f(0), \tag{1.65}$$

由定义立即得到

$$\int_{-\infty}^{\infty}\delta(x-a)\,f(x)\,\mathrm{d}x=f(a). \tag{1.66}$$

设 $g(x)$ 在 x_i 处有孤立零点，且 $g'(x_i)\neq 0$，则

$$\delta(g(x))=\sum_i\frac{\delta(x-x_i)}{|g'(x_i)|}. \tag{1.67}$$

1.5.2 阶跃函数

定义阶跃函数

$$H(x)=\begin{cases}0, & x<0,\\1, & x>0.\end{cases} \tag{1.68}$$

1.6 阶跃函数的导数

在通常意义下，$H(x)$ 在 $x=0$ 不可导。但在积分意义下，我们定义

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}H(x)}{\mathrm{d}x} f(x)\ \mathrm{d}x \equiv -\int_{-\infty}^{\infty} H(x) \frac{\mathrm{d}f}{\mathrm{d}x}\ \mathrm{d}x. \tag{1.69}$$

对右侧分部积分：

$$\begin{aligned} -\int_{-\infty}^{\infty} H(x)\frac{\mathrm{d}f}{\mathrm{d}x}\,\mathrm{d}x &= -\int_{0}^{\infty} \frac{\mathrm{d}f}{\mathrm{d}x}\,\mathrm{d}x \\ &= f(0). \end{aligned} \tag{1.70}$$

于是得到

$$\frac{\mathrm{d}H(x)}{\mathrm{d}x} = \delta(x), \tag{1.71}$$

1.6.1 多维狄拉克函数

三维 δ 函数定义为

$$\delta^{(3)}(\boldsymbol{r}) = \delta(x)\delta(y)\delta(z), \tag{1.72}$$

并满足

$$\iiint \delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_0) f(\boldsymbol{r})\ \mathrm{d}V = f(\boldsymbol{r}_0). \tag{1.73}$$

在正交曲线坐标 (u_1, u_2, u_3) 中，

$$\mathrm{d}V = h_1 h_2 h_3\ \mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3,$$

因此定义

$$\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_0) = \frac{\delta(u_1-u_{1,0})\delta(u_2-u_{2,0})\delta(u_3-u_{3,0})}{h_1 h_2 h_3}. \tag{1.74}$$

1.7 矢量场理论

1.7.1 亥姆霍兹定理

设矢量场 $\boldsymbol{F}(\boldsymbol{r})$ 定义在整个空间中，其散度与旋度分别给定为

$$\boldsymbol{\nabla}\cdot\boldsymbol{F}(\boldsymbol{r}) = D(\boldsymbol{r}), \qquad \boldsymbol{\nabla}\times\boldsymbol{F}(\boldsymbol{r}) = \boldsymbol{C}(\boldsymbol{r}).$$

为了使体积分有意义，仅要求 $D(\boldsymbol{r})$ 与 $\boldsymbol{C}(\boldsymbol{r})$ 在无穷远处趋于零是不够的，它们的衰减速度必须足够快。

考虑定义在整个空间上的积分

$$\iiint_{\mathbb{V}} X(\boldsymbol{r})\,\mathrm{d}V,$$

其中 $X(\boldsymbol{r})$ 表示 $D(\boldsymbol{r})$ 或 $\boldsymbol{C}(\boldsymbol{r})$ 的某一分量。

将积分写为球坐标形式，

$$\iiint_{\mathbb{V}} X(\boldsymbol{r})\,\mathrm{d}V = \int_0^{\infty}\left(\iint_S X(r,\Omega) r^2\,\mathrm{d}\Omega\right)\mathrm{d}r.$$

若存在常数 $A>0$ 及 $R>0$，使得当 $r>R$ 时对所有方向 Ω 成立

$$|X(r,\Omega)| \geq \frac{A}{r},$$

则有下界估计

$$\iiint_{|\boldsymbol{r}|>R} |X(\boldsymbol{r})|\,\mathrm{d}V \geq A\int_R^\infty r\,\mathrm{d}r \iint_S \mathrm{d}\Omega,$$

该积分显然发散。

类似地，若

$$|X(r,\Omega)| \geq \frac{A}{r^2},$$

则径向积分包含

$$\int_R^\infty \frac{\mathrm{d}r}{r},$$

从而发散。

因此，为保证体积分收敛，必须要求

$$X(\boldsymbol{r}) = o\left(\frac{1}{r^2}\right), \qquad r\to\infty, \tag{1.75}$$

该条件需在角向上一致成立。

因此，为保证积分收敛，必须要求

$$D(\boldsymbol{r}),\ \boldsymbol{C}(\boldsymbol{r}) = o\left(\frac{1}{r^2}\right), \qquad r\to\infty. \tag{1.76}$$

这一条件同时也足以保证在无穷远处的曲面积分为零。

设矢量场 $\boldsymbol{F}(\boldsymbol{r})$ 具有散度 $D(\boldsymbol{r})$ 与旋度 $\boldsymbol{C}(\boldsymbol{r})$。若取另一矢量场

$$\boldsymbol{F}'(\boldsymbol{r}) = \boldsymbol{F}(\boldsymbol{r}) + \boldsymbol{G}(\boldsymbol{r}),$$

且 $\boldsymbol{G}(\boldsymbol{r})$ 满足

$$\boldsymbol{\nabla}\cdot\boldsymbol{G} = 0, \qquad \boldsymbol{\nabla}\times\boldsymbol{G} = \boldsymbol{0},$$

则 $\boldsymbol{F}'$ 与 $\boldsymbol{F}$ 具有相同的散度与旋度。

因此，仅给定散度与旋度并不能唯一确定矢量场。然而可以证明：不存在一个非零的矢量场 (见第三章) $\boldsymbol{G}(\boldsymbol{r})$，它在整个空间中同时满足

$$\boldsymbol{\nabla}\cdot\boldsymbol{G} = 0, \qquad \boldsymbol{\nabla}\times\boldsymbol{G} = \boldsymbol{0},$$

并且在无穷远处趋于零。

因此，若进一步要求

$$\boldsymbol{F}(\boldsymbol{r}) \to \boldsymbol{0} \qquad 当 r\to\infty,$$

则满足给定散度与旋度的矢量场解是唯一的。

现在可以严谨地表述亥姆霍兹定理如下：

> 若矢量场 $\boldsymbol{F}(\boldsymbol{r})$ 的散度 $D(\boldsymbol{r})$ 与旋度 $\boldsymbol{C}(\boldsymbol{r})$ 已知，且二者在 $r\to\infty$ 时均比 $1/r^2$ 衰减得更快，同时 $\boldsymbol{F}(\boldsymbol{r})$ 在无穷远处趋于零，则 $\boldsymbol{F}(\boldsymbol{r})$ 被其散度与旋度唯一确定。

1.7.2 势函数

1. 标量势与无旋场 设向量场 $\boldsymbol{F}(\boldsymbol{r})$ 定义在某一区域 $V\subset\mathbb{R}^3$。若存在标量函数 $\phi(\boldsymbol{r})$，使得

$$\boldsymbol{F} = -\boldsymbol{\nabla}\phi, \tag{1.77}$$

则称 ϕ 为 $\boldsymbol{F}$ 的**标量势函数**，$\boldsymbol{F}$ 称为**保守场**。由旋度的定义可得

$$\boldsymbol{\nabla}\times\boldsymbol{F} = -\boldsymbol{\nabla}\times\boldsymbol{\nabla}\phi = \boldsymbol{0}. \tag{1.78}$$

因此，任何具有标量势的向量场必为无旋场。

2. 无旋场的势函数存在性 若

$$\boldsymbol{\nabla} \times \boldsymbol{F} = \boldsymbol{0}, \tag{1.79}$$

并且区域 V 是单连通的（任意闭合曲线可连续收缩为一点），则由斯托克斯定理可得

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = \iint_S (\boldsymbol{\nabla} \times \boldsymbol{F}) \cdot \mathrm{d}\boldsymbol{S} = 0, \tag{1.80}$$

线积分与路径无关，可定义

$$\phi(\boldsymbol{r}) = -\int_{\boldsymbol{r}_0}^{\boldsymbol{r}} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l}, \tag{1.81}$$

从而得到 $\boldsymbol{F} = -\boldsymbol{\nabla}\phi$。

3. 向量势与无散场 若存在向量函数 $\boldsymbol{A}(\boldsymbol{r})$，使

$$\boldsymbol{F} = \boldsymbol{\nabla} \times \boldsymbol{A}, \tag{1.82}$$

则称 $\boldsymbol{A}$ 为 $\boldsymbol{F}$ 的**向量势**。由恒等式

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = 0 \tag{1.83}$$

可知，任何具有向量势的向量场必为无散场。

4. 势函数的不唯一性

- 若 $\boldsymbol{F} = -\boldsymbol{\nabla}\phi$，则对任意常数 C，$\phi' = \phi + C$ 给出同一向量场。
- 若 $\boldsymbol{F} = \boldsymbol{\nabla} \times \boldsymbol{A}$，则对任意标量函数 $\chi(\boldsymbol{r})$，$\boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\nabla}\chi$ 产生相同的 $\boldsymbol{F}$（**规范变换**）。

5. 亥姆霍兹分解 在满足适当衰减条件的情况下，任意向量场 $\boldsymbol{F}$ 可分解为

$$\boldsymbol{F} = -\boldsymbol{\nabla}\phi + \boldsymbol{\nabla} \times \boldsymbol{A}, \tag{1.84}$$

其中

$$\boldsymbol{\nabla}^2\phi = -\boldsymbol{\nabla} \cdot \boldsymbol{F}, \tag{1.85}$$

$$\boldsymbol{\nabla}^2\boldsymbol{A} = -\boldsymbol{\nabla} \times \boldsymbol{F}. \tag{1.86}$$

这就是亥姆霍兹定理在势函数语言下的表达。

1.8 习题

1.8.1 证明: $\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) + \boldsymbol{c} \times (\boldsymbol{a} \times \boldsymbol{b}) + \boldsymbol{b} \times (\boldsymbol{c} \times \boldsymbol{a}) = 0$

$$\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})+\boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b})+\boldsymbol{b}\times(\boldsymbol{c}\times\boldsymbol{a}) = -(\boldsymbol{b}\cdot\boldsymbol{a})\boldsymbol{c}+(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c}-(\boldsymbol{a}\cdot\boldsymbol{c})\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{c})\boldsymbol{a}-(\boldsymbol{c}\cdot\boldsymbol{b})\boldsymbol{a}+(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c} = 0$$

1.8.2 证明: $\boldsymbol{Ra} \cdot \boldsymbol{Rb} = \boldsymbol{a} \cdot \boldsymbol{b}$, 其中 $\boldsymbol{R}$ 为旋转矩阵

$$\boldsymbol{Ra} \cdot \boldsymbol{Rb} = (\boldsymbol{Ra})^T(\boldsymbol{Rb}) = \boldsymbol{a}^T\boldsymbol{R}^T\boldsymbol{Rb} = \boldsymbol{a}^T\boldsymbol{R}^{-1}\boldsymbol{Rb} = \boldsymbol{a}^T\boldsymbol{b} = \boldsymbol{a} \cdot \boldsymbol{b}$$

1.8.3 在坐标逆变换 $(x'=-x,\ y'=-y,\ z'=-z)$ 下, 两个矢量的叉乘是如何变换的?

$$\boldsymbol{a}'\times\boldsymbol{b}'=\begin{pmatrix}\boldsymbol{e}_i & \boldsymbol{e}_j & \boldsymbol{e}_k\\ a'_i & a'_j & a'_k\\ b'_i & b'_j & b'_k\end{pmatrix}=\begin{pmatrix}\boldsymbol{e}_i & \boldsymbol{e}_j & \boldsymbol{e}_k\\ -a_i & -a_j & -a_k\\ -b_i & -b_j & -b_k\end{pmatrix}=\boldsymbol{a}\times\boldsymbol{b}$$

1.8.4 在坐标逆变换下, 标量三重积 $(\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b}))$ 是如何变换的?

$$\boldsymbol{c}'\cdot(\boldsymbol{a}'\times\boldsymbol{b}')=\boldsymbol{c}'\cdot(\boldsymbol{a}\times\boldsymbol{b})=-\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b})$$

1.8.5 $\boldsymbol{n}$ 是固定点 (x',y',z') 到点 (x,y,z) 的间隔矢量, n 是它的长度, 证明:

(a) $\boldsymbol{\nabla}n^2=2\boldsymbol{n}$.

(b) $\boldsymbol{\nabla}\frac{1}{n}=-\frac{\boldsymbol{n}}{n^2}$.

(c) $\boldsymbol{\nabla}n^a=a\boldsymbol{n}n^{a-2}(a\neq 0)$.

(a) $\boldsymbol{\nabla}n^2=\frac{\partial n^2}{\partial e_i}\boldsymbol{e}_i=2n\frac{\partial n}{\partial e_i}\boldsymbol{e}_i=2\boldsymbol{n}$

(c) $\boldsymbol{\nabla}n^a=\frac{\partial n^a}{\partial e_i}\boldsymbol{e}_i=an^{-1}\frac{\partial n}{\partial e_i}\boldsymbol{e}_i=a\boldsymbol{n}n^{a-2}$

1.8.6 函数 f 只依赖平面坐标 y,z。令旋转后的坐标记为 $\begin{pmatrix}y'\\ z'\end{pmatrix}=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}y\\ z\end{pmatrix}$, 求证 $\boldsymbol{\nabla}f'=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\frac{\partial f}{\partial y}\\ \frac{\partial f}{\partial z}\end{pmatrix}$

$$\begin{aligned}
\begin{pmatrix}y\\ z\end{pmatrix}&=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}y'\\ z'\end{pmatrix}\\
\frac{\partial f}{\partial y'}&=\frac{\partial f}{\partial y}\frac{\partial y}{\partial y'}+\frac{\partial f}{\partial z}\frac{\partial z}{\partial y'}=\frac{\partial f}{\partial y}\cos\theta+\frac{\partial f}{\partial z}\sin\theta\\
\frac{\partial f}{\partial z'}&=\frac{\partial f}{\partial y}\frac{\partial y}{\partial z'}+\frac{\partial f}{\partial z}\frac{\partial z}{\partial z'}=-\frac{\partial f}{\partial y}\sin\theta+\frac{\partial f}{\partial z}\cos\theta\\
\begin{pmatrix}\dfrac{\partial f}{\partial y'}\\ \dfrac{\partial f}{\partial z'}\end{pmatrix}&=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\dfrac{\partial f}{\partial y}\\ \dfrac{\partial f}{\partial z}\end{pmatrix}
\end{aligned}$$

1.8.7 计算 $(\boldsymbol{\nabla T})\times(\boldsymbol{\nabla S})$

$$
\begin{aligned}
(\boldsymbol{\nabla T})\times(\boldsymbol{\nabla S}) &= \left(\varepsilon_{ijk}\frac{\partial T_j}{\partial x_i}\boldsymbol{e}_k\right)\times\left(\varepsilon_{lmn}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_n\right)\\
&= \varepsilon_{opq}\varepsilon_{ijo}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= \varepsilon_{pqo}\varepsilon_{ijo}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= (\delta_{ip}\delta_{jq}-\delta_{iq}\delta_{jp})\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= \delta_{ip}\delta_{jq}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q-\delta_{iq}\delta_{jp}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= \frac{\partial T_j}{\partial x_i}\varepsilon_{lmi}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_j-\frac{\partial T_j}{\partial x_i}\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i\\
&= \frac{\partial T_i}{\partial x_j}\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i-\frac{\partial T_j}{\partial x_i}\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i\\
&= \left(\frac{\partial T_i}{\partial x_j}-\frac{\partial T_j}{\partial x_i}\right)\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i
\end{aligned}
$$

1.8.8 证明 $\iint_S f(\boldsymbol{\nabla}\times\boldsymbol{a})\cdot\mathrm{d}\boldsymbol{S}=\iint_S[\boldsymbol{a}\times(\boldsymbol{\nabla}f)]\cdot\mathrm{d}\boldsymbol{S}+\oint_{\partial S}f\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}$

$$
\begin{aligned}
\oint_{\partial S}f\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l} &= \iint_S[\boldsymbol{\nabla}\times(\boldsymbol{f}\boldsymbol{a})]\cdot\mathrm{d}\boldsymbol{S}\\
&= \iint_S[f(\boldsymbol{\nabla}\times\boldsymbol{a})+(\boldsymbol{\nabla}f)\times\boldsymbol{a}]\cdot\mathrm{d}\boldsymbol{S}\\
\iint_S f(\boldsymbol{\nabla}\times\boldsymbol{a})\cdot\mathrm{d}\boldsymbol{S} &= \iint_S[\boldsymbol{a}\times(\boldsymbol{\nabla}f)]\cdot\mathrm{d}\boldsymbol{S}+\oint_{\partial S}f\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}
\end{aligned}
$$

1.8.9 证明 $\iiint_V\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}\tau=\iiint_V\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})\,\mathrm{d}\tau+\iint(\boldsymbol{a}\times\boldsymbol{b})\cdot\mathrm{d}\boldsymbol{S}_{\partial V}$

$$
\begin{aligned}
\iint_{\partial V}(\boldsymbol{a}\times\boldsymbol{b})\cdot\mathrm{d}\boldsymbol{S} &= \iiint_V\boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{b})\,\mathrm{d}V\\
&= \iiint_V[\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})-\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})]\,\mathrm{d}V\\
\iiint_V\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}\tau &= \iiint_V\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})\,\mathrm{d}\tau+\iint_{\partial V}(\boldsymbol{a}\times\boldsymbol{b})\cdot\mathrm{d}\boldsymbol{S}
\end{aligned}
$$

1.8.10 设 $f=f(r)$, 求 $\boldsymbol{\nabla}^2[f(r)]$

$$
\begin{aligned}
\boldsymbol{\nabla}^2[f(r)] &= \frac{1}{sr}\frac{\partial}{\partial r}\left(sr\frac{\partial f}{\partial r}\right)+\frac{1}{s^2}\frac{\partial}{\partial \phi}\left(\frac{\partial f}{\partial \phi}\right)+\frac{1}{sr^2}\frac{\partial}{\partial \theta}\left(s\frac{\partial f}{\partial \theta}\right)\\
&= \frac{1}{sr}\frac{\partial}{\partial r}\left(sr\frac{\partial f}{\partial r}\right)\\
&= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right)\\
&= \frac{2}{r}\frac{\partial f}{\partial r}+\frac{\partial^2 f}{\partial r^2}
\end{aligned}
\tag{1.87}
$$

1.8.11 证明:$\iiint_V \boldsymbol{\nabla}\times\boldsymbol{a}\,\mathrm{d}V=\oiint \mathrm{d}\boldsymbol{S}\times\boldsymbol{a}$

取任意常向量 $\boldsymbol{c}$。利用式1.39得 $\boldsymbol{\nabla}\times\boldsymbol{c}=\boldsymbol{0}$, 因此

$$\boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{c})=\boldsymbol{c}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a}).$$

应用高斯定理,

$$\iiint_V \boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{c})\,\mathrm{d}V=\oiint_{\partial V}(\boldsymbol{a}\times\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{S}=\iiint_V \boldsymbol{c}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}V.$$

利用式1.12得

$$\boldsymbol{c}\cdot\iiint_V(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}V=\boldsymbol{c}\cdot\oiint_{\partial V}\mathrm{d}\boldsymbol{S}\times\boldsymbol{a}.$$

由于上述等式对任意常向量 $\boldsymbol{c}$ 成立, 必有

$$\iiint_V(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}V=\oiint_{\partial V}\mathrm{d}\boldsymbol{S}\times\boldsymbol{a}.$$

1.8.12 证明:$\iint_S \mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f=\oint_{\partial S} f\,\mathrm{d}\boldsymbol{l}$

取任意常向量 $\boldsymbol{c}$。利用式1.28得

$$\iint_S(\boldsymbol{\nabla}f\times\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{S}=\iint_S(\boldsymbol{\nabla}\times(f\boldsymbol{c}))\cdot\mathrm{d}\boldsymbol{S}.$$

由斯托克斯定理

$$\iint_S(\boldsymbol{\nabla}\times\boldsymbol{A})\cdot\mathrm{d}\boldsymbol{S}=\oint_{\partial S}\boldsymbol{A}\cdot\mathrm{d}\boldsymbol{l},$$

取 $\boldsymbol{A}=f\boldsymbol{c}$，得

$$\iint_S(\boldsymbol{\nabla}\times(f\boldsymbol{c}))\cdot\mathrm{d}\boldsymbol{S}=\oint_{\partial S}(f\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{l}=\oint_{\partial S}f\,\boldsymbol{c}\cdot\mathrm{d}\boldsymbol{l}.$$

又由混合积恒等式

$$\boldsymbol{c}\cdot(\mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f)=(\boldsymbol{\nabla}f\times\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{S},$$

所以

$$\boldsymbol{c}\cdot\iint_S\mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f=\boldsymbol{c}\cdot\oint_{\partial S}f\ \mathrm{d}\boldsymbol{l}.$$

由于上述等式对任意常向量 $\boldsymbol{c}$ 都成立，必有

$$\iint_S\mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f=\oint_{\partial S}f\ \mathrm{d}\boldsymbol{l}.$$

1.8.13 证明:$\oiint_{\partial S}(\boldsymbol{a}\times\boldsymbol{r})\cdot\mathrm{d}\boldsymbol{l}=2\iint_S\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S}$

$$
\begin{aligned}
\oiint_{\partial S}(\boldsymbol{a}\times\boldsymbol{r})\cdot\mathrm{d}\boldsymbol{l}&=\iint_S[\boldsymbol{\nabla}\times(\boldsymbol{a}\times\boldsymbol{r})]\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S[\boldsymbol{\nabla}\times(\varepsilon_{ijk}a_ix_j\boldsymbol{e}_k)]\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\varepsilon_{lmn}\varepsilon_{ijm}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=-\iint_S(\varepsilon_{lnm}\varepsilon_{ijm}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=-\iint_S(\delta_{il}\delta_{jn}-\delta_{in}\delta_{jl})(\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\delta_{in}\delta_{jl}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n-\delta_{il}\delta_{jn}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\frac{\partial a_ix_j}{\partial x_j}\boldsymbol{e}_i-\frac{\partial a_ix_j}{\partial x_i}\boldsymbol{e}_j)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\frac{\partial a_jx_i}{\partial x_i}\boldsymbol{e}_j-\frac{\partial a_ix_j}{\partial x_i}\boldsymbol{e}_j)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(a_j\frac{\partial x_i}{\partial x_i}+x_i\frac{\partial a_j}{\partial x_i}-a_i\frac{\partial x_j}{\partial x_i}-x_j\frac{\partial a_i}{\partial x_i})\boldsymbol{e}_j\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(3a_j+x_i\frac{\partial a_j}{\partial x_i}-a_i\delta_{ij}-x_j\frac{\partial a_i}{\partial x_i})\boldsymbol{e}_j\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S 2a_j\boldsymbol{e}_j\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S}
\end{aligned}
$$

1.8.14 将下列电荷分布表示成电荷密度函数 ρ:1. 电荷量 Q 均匀分布在半径为 R 的球面上; 2. 电荷均匀分布在半径为 R 的柱面上, 单位长度的电荷量为 λ; 3. 电荷量均匀分布在半径为 R 的平面圆盘上

$$
\begin{aligned}
&1.\rho=\delta(R\boldsymbol{e}_r-\boldsymbol{r})\frac{Q}{4\pi R^2}\\
&2.\rho=\delta(R\boldsymbol{e}_s-\boldsymbol{s})\frac{\lambda}{2\pi R}\\
&3.\rho=\delta(\boldsymbol{x}_3)H(x+R)H(-x-R)\frac{\lambda}{\pi R^2}
\end{aligned}
$$

1.8.15 对函数 $\boldsymbol{a} = r^2\cos\theta\boldsymbol{r} + r^2\cos\phi\boldsymbol{\theta} - r^2\cos\theta\sin\phi\boldsymbol{\phi}$ 验证散度定理, 体积为半径为 R 在第一卦限的 $1/8$ 球体

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = 2r\cos\theta - r^2\cos\theta\cos\phi$$

$$\begin{aligned}
\int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R\boldsymbol{\nabla}\cdot\boldsymbol{a}\,\mathrm{d}r &= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R\left(2r\cos\theta - r^2\cos\theta\cos\phi\right)\mathrm{d}r \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\left(r^2\cos\theta - \frac{r^3}{3}\cos\theta\cos\phi\right)\Big|_0^R \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\left(R^2\cos\theta - \frac{R^3}{3}\cos\theta\cos\phi\right)\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\left(R^2\sin\theta - \frac{R^3}{3}\sin\theta\cos\phi\right)\Big|_0^{\frac{\pi}{2}} \\
&= \int_0^{\frac{\pi}{2}}\left(R^2 - \frac{R^3}{3}\cos\phi\right)\mathrm{d}\phi \\
&= \left(R^2\phi - \frac{R^3}{3}\sin\phi\right)\Big|_0^{\frac{\pi}{2}} \\
&= \frac{\pi}{2}R^2 - \frac{R^3}{3}
\end{aligned}$$

计算 XY 平面:

$$\begin{aligned}
\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R a_\phi\,\mathrm{d}r &= -\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R r^2\cos\theta\sin\phi\,\mathrm{d}r \\
&= -\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R r^2\cos\theta\,\mathrm{d}r \\
&= -\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\,\frac{r^3}{3}\cos\theta\Big|_0^R \\
&= -\int_0^{\frac{\pi}{2}}\frac{R^3}{3}\cos\theta\,\mathrm{d}\theta \\
&= -\frac{R^3}{3}
\end{aligned}$$

计算 XZ 平面:

$$\begin{aligned}
\int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R a_\theta\,\mathrm{d}r &= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R r^2\cos\phi\,\mathrm{d}r \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\,\frac{r^3}{3}\cos\phi\Big|_0^R \\
&= \int_0^{\frac{\pi}{2}}\frac{R^3}{3}\cos\phi\,\mathrm{d}\phi \\
&= \frac{R^3}{3}
\end{aligned}$$

计算 YZ 平面:

$$\int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R a_\theta\,\mathrm{d}r = \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R r^2\cos\phi\,\mathrm{d}r = \frac{R^3}{3}$$

1.8.16 对函数 $\boldsymbol{a}=bx_2\boldsymbol{e}_1+cx_1\boldsymbol{e}_2$ 验证斯托克斯定理, 面的边界线选为处在 XY 平面, 半径为 R, 圆心在原点的圆周线。

$$\boldsymbol{\nabla}\times\boldsymbol{a}=\boldsymbol{\nabla}\times(by\boldsymbol{e}_1+cx\boldsymbol{e}_2)=\begin{pmatrix}\boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3\\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3}\\ bx_2 & cx_1 & 0\end{pmatrix}=(c-b)\boldsymbol{e}_3$$

$$\iint\boldsymbol{\nabla}\times\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}=\iint(c-b)\boldsymbol{e}_3\cdot\mathrm{d}\boldsymbol{S}=2\pi R(c-b)$$

$$\begin{aligned}\oint(bR\sin\phi\boldsymbol{e}_1+cR\cos\phi\boldsymbol{e}_2)\cdot\mathrm{d}\boldsymbol{l}=&\int_0^{2\pi}(bR\sin\phi\boldsymbol{e}_1+cR\cos\phi\boldsymbol{e}_2)\cdot(-R\sin\phi\boldsymbol{e}_1\ +R\cos\phi\boldsymbol{e}_2\ )\,\mathrm{d}\phi\\ =&\int_0^{2\pi}(-bR^2\sin^2\phi+cR^2\cos^2\phi)\,\mathrm{d}\phi\\ =&2\pi R(c-b)\end{aligned}$$

1.8.17 对函数 $\boldsymbol{a}=r\cos^2\theta\boldsymbol{r}-r\sin\theta\cos\theta\boldsymbol{\theta}+3r\boldsymbol{\phi}$ 验证斯托克斯定理。

$$\begin{aligned}&\boldsymbol{\nabla}\times\boldsymbol{a}\\ =&\frac{1}{sr}\left(r\frac{\partial a_\theta}{\partial\phi}-s\frac{\partial a_\phi}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial a_r}{\partial\theta}-\frac{\partial ra_\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{s}\left(\frac{\partial sa_\phi}{\partial r}-\frac{\partial a_r}{\partial\phi}\right)\boldsymbol{\theta}\\ =&\frac{1}{sr}\left(r\frac{\partial r\sin\theta\cos\theta}{\partial\phi}-s\frac{\partial 3r}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial r\cos^2\theta}{\partial\theta}-\frac{\partial rr\sin\theta\cos\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{s}\left(\frac{\partial s3r}{\partial r}-\frac{\partial r\cos^2\theta}{\partial\phi}\right)\boldsymbol{\theta}\\ =&\frac{1}{s}\left(6r\sin\theta\right)\boldsymbol{\theta}\\ =&6\boldsymbol{\theta}\end{aligned}$$

计算 XY 平面:

$$\begin{aligned}\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R a_\phi\,\mathrm{d}r=&\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R 6\,\mathrm{d}r\\ =&\frac{3}{2}R\end{aligned}$$

1.8.18 对函数 $\boldsymbol{a} = r^2\sin^2\theta\boldsymbol{r} + 4r^2\cos\theta\boldsymbol{\theta} + r^2\tan\theta\boldsymbol{\phi}$ 验证散度定理。

$$
\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{a} &= \frac{1}{sr}(\frac{\partial sra_r}{\partial r} + r\frac{\partial a_\phi}{\partial\phi} + s\frac{\partial\theta}{\partial\theta}) \\
&= \frac{1}{sr}(\frac{\partial srr^2\sin^2\theta}{\partial r} + r\frac{\partial 4r^2\cos\theta}{\partial\phi} + s\frac{\partial r^2\tan\theta}{\partial\theta}) \\
&= \frac{1}{sr}(4sr^2\sin^2\theta + sr^2\frac{1}{\cos^2\theta}) \\
&= 4r\sin^2\theta + r\frac{1}{\cos^2\theta}
\end{aligned}
$$

$$
\begin{aligned}
&\int_0^R r^2\,\mathrm{d}r\int_0^{2\pi}\mathrm{d}\phi\int_0^{\frac{\pi}{6}}(4r\sin^2\theta + r\frac{1}{\cos^2\theta})\sin\theta\,\mathrm{d}\theta \\
=&\int_0^R 2\pi r^3\,\mathrm{d}r\int_0^{\frac{\pi}{6}}(4\sin^2\theta + \frac{1}{\cos^2\theta})\,\mathrm{d}\theta \\
=&\int_0^R 2\pi r^3\ (2\theta - \sin 2\theta + \tan\theta)|_0^{\frac{\pi}{6}}\ \mathrm{d}r \\
=&\int_0^R 2\pi r^3(\frac{\pi}{3} - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2})\,\mathrm{d}r \\
=&\int_0^R 2\pi r^3\frac{\pi}{3}\,\mathrm{d}r \\
=&2\pi R^4\frac{\pi}{12}
\end{aligned}
$$

1.8.19 证明 $\iiint_V\left[f\boldsymbol{\nabla}^2 g + (\boldsymbol{\nabla}f)\cdot(\boldsymbol{\nabla}g)\right]\mathrm{d}V = \oiint_S(f\boldsymbol{\nabla}g)\cdot\mathrm{d}S$

$$
\begin{aligned}
\iiint_V(\boldsymbol{\nabla}\cdot\boldsymbol{a})\,\mathrm{d}V &= \iint_{\partial V}\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S} \\
\iiint_V[\boldsymbol{\nabla}\cdot(f\boldsymbol{\nabla}g)]\,\mathrm{d}V &= \iint_{\partial V}(f\boldsymbol{\nabla}g)\cdot\mathrm{d}\boldsymbol{S} \\
\iiint_V\left[f\boldsymbol{\nabla}^2 g + (\boldsymbol{\nabla}f)\cdot(\boldsymbol{\nabla}g)\right]\mathrm{d}V &= \iint_{\partial V}(f\boldsymbol{\nabla}g)\cdot\mathrm{d}\boldsymbol{S}
\end{aligned}
$$

1.8.20 求 $\boldsymbol{a} = r^n \boldsymbol{e}_r$ 的散度

$$
\begin{aligned}
\boldsymbol{\nabla} \cdot \boldsymbol{a} &= \frac{\partial a_i}{\partial x_i} \\
&= \frac{\partial x_i r^{n-1}}{\partial x_i} \\
&= 3r^{n-1} + x_i(n-1)r^{n-2}\frac{\partial r}{\partial x_i} \\
&= 3r^{n-1} + x_i(n-1)r^{n-2}\frac{x_i}{r} \\
&= 3r^{n-1} + (n-1)r^{n-1} \\
&= (n+2)r^{n-1}
\end{aligned}
$$

$$
\iint_{\partial V} \boldsymbol{a} \cdot \mathrm{d}\boldsymbol{S} = \iint_{\partial V} r^n \boldsymbol{e}_r \cdot \mathrm{d}\boldsymbol{S} = 4\pi r^2 r^n =
$$

1.8.21 求 $\boldsymbol{a} = r^n \boldsymbol{e}_r$ 的旋度

$$
\begin{aligned}
\boldsymbol{\nabla} \times \boldsymbol{a} &= \varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_k \\
&= \varepsilon_{ijk}\frac{\partial r^{n-1}x_j}{\partial x_i}\boldsymbol{e}_k \\
&= \varepsilon_{ijk}x_j\frac{\partial r^{n-1}}{\partial x_i}\boldsymbol{e}_k \\
&= (n-1)\varepsilon_{ijk}x_j r^{n-2}\frac{\partial r}{\partial x_i}\boldsymbol{e}_k \\
&= (n-1)\varepsilon_{ijk}x_j r^{n-2}\frac{x_i}{r}\boldsymbol{e}_k \\
&= (n-1)r^{n-3}\varepsilon_{ijk}x_i x_j \boldsymbol{e}_k \\
&= 0
\end{aligned}
$$

2 静电学

2.1 电场

2.1.1 库仑定律

设一个静止点电荷 q 距检验电荷 Q 的距离为 r, 那么它作用在 Q 上的力是

$$\boldsymbol{F} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^3}\boldsymbol{r} \tag{2.1}$$

常数 ε_0 称为真空介电常数, $\varepsilon_0 = 8.85 \times 10^{-12}\ \dfrac{\mathrm{C}^2}{\mathrm{N}\cdot\mathrm{m}^2}$

2.1.2 电场

$$\boldsymbol{E} \equiv \frac{\boldsymbol{F}}{Q} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r} \tag{2.2}$$

2.1.3 电场强度通量

$\boldsymbol{\Phi}_E$ 为电场强度通量

$$\boldsymbol{\Phi}_E \equiv \iint_S \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{S} \tag{2.3}$$

2.1.4 高斯定理

ρ 为电荷密度

$$\boldsymbol{\nabla}\cdot\boldsymbol{E} = \frac{1}{\varepsilon_0}\rho \tag{2.4}$$

2.1.5 两个带点为 q 的电荷, 相距 d 放置, 求垂直于连线中点且距离为 x_2 处的电场, 如果把一个换成 $-q$ 会怎样? 当 $y >> d$ 时会怎样?

$$\begin{aligned}\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_1 + \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_2 &= 2\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}x_2\boldsymbol{e}_2 \\ &= \frac{1}{2\pi\varepsilon_0}\frac{qx_2}{r^3}\boldsymbol{e}_2 \\ &= \frac{1}{2\pi\varepsilon_0}\frac{qx_2}{(x_2^2+\frac{d^2}{4})^{\frac{3}{2}}}\boldsymbol{e}_2\end{aligned}$$

$$\begin{aligned}\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_1 + \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_2 &= 2\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\frac{d}{2}\boldsymbol{e}_1 \\ &= \frac{1}{4\pi\varepsilon_0}\frac{qd}{r^3}\boldsymbol{e}_1 \\ &\approx \frac{1}{4\pi\varepsilon_0}\frac{qd}{x_2^3}\boldsymbol{e}_1\end{aligned}$$

2.1.6 一个长度为 $2L$ 的细杆均匀带电, 电荷线密度为 λ, 求垂直于杆且与杆中心距离为 x_2 处的电场

$$
\begin{aligned}
\int_{-L}^{L} \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{r^3}\boldsymbol{r}\,\mathrm{d}x &= \int_{-L}^{L} \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{r^3}x_2\boldsymbol{e}_2\,\mathrm{d}x \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-L}^{L}\frac{x_2}{r^3}\,\mathrm{d}x \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-L}^{L}\frac{x_2}{(x^2+x_2^2)^{\frac{3}{2}}}\,\mathrm{d}x \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\frac{2L}{x_2\sqrt{L^2+x_2^2}}
\end{aligned}
$$

2.1.7 一个长度为 L 的细杆均匀带电, 电荷线密度为 λ, 求杆一端上方距杆 y 处的电场

$$
\begin{aligned}
\int_0^L \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0 r^2}\,\mathrm{d}x_2 &= \int_0^L \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0(x_2+y)^2}\,\mathrm{d}x_2 \\
&= -\left.\frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0(x_2+y)}\right|_0^L \\
&= \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0(L+y)} - \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0 y} \\
&= \frac{\lambda\boldsymbol{e}_y L}{4\pi\varepsilon_0(L+y)y}
\end{aligned}
$$

2.1.8 一个边长为 L 的正方形线框均匀带电, 电荷线密度为 λ, 求线框中心上方距 x_3 处的电场

$$
4\frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_3\frac{Lx_3}{x_3\sqrt{2\left(\frac{L}{2}\right)^2+x_3^2}^2} = \frac{\lambda}{\pi\varepsilon_0}\boldsymbol{e}_3\frac{L}{\frac{L^2}{2}+x_3^2}
$$

2.1.9 一个半径为 r 的圆线框均匀带电, 电荷线密度为 λ, 求线框中心上方距 x_3 处的电场

$$
2\pi r\frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+r^2}} = \frac{\lambda}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3 r}{\sqrt{x_3^2+r^2}^3}
$$

2.1.10 一个半径为 r 的圆片均匀带电, 电荷面密度为 σ, 求圆片中心上方距 x_3 处的电场

$$
\begin{aligned}
\int_0^r \frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3x_1}{\sqrt{x_3^2+x_1^2}^3}\,\mathrm{d}x_1 &= -\left.\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+x_1^2}}\right|_0^r \\
&= \frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2}} - \frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+r^2}} \\
&= \frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3 - \frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+r^2}}
\end{aligned}
$$

2.1.11 一个半径为 r 的球面均匀带电, 电荷面密度为 σ, 求圆球上方距 d 处的电场 (分 $d>r$ 和 $d<r$ 讨论)

$$
\begin{aligned}
\int_{-r}^r \frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(x_3+d)\sqrt{x_1^2+x_2^2}}{\sqrt{(x_3+d)^2+x_1^2+x_2^2}^3}\,\mathrm{d}x_3 &= \int_\pi^0 \frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(r\cos\phi+d)\sqrt{x_1^2+x_2^2}}{\sqrt{(r\cos\phi+d)^2+x_1^2+x_2^2}^3}\,\mathrm{d}r\cos\phi \\
&= \int_\pi^0 \frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(r\cos\phi+d)r\sin\phi}{\sqrt{r^2+2rd\cos\phi+d^2}^3}\,\mathrm{d}r\cos\phi \\
&= \int_0^\pi \frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(r\cos\phi+d)r^2\sin^2\phi}{\sqrt{r^2+2rd\cos\phi+d^2}^3}\,\mathrm{d}\phi
\end{aligned}
$$